

3.2

a) span $\{1, x, x^2\}$, oznaime $v_1 = 1, v_2 = x, v_3 = x^2$
pomocí gramovy 6-5 or ortonormalizace ~~ne~~ vytvoříme
ortonormální bázi $B = (u_1, u_2, u_3)$

$$u_1 = \frac{1}{\sqrt{\int_1^5 1 dx}} = \frac{1}{2}$$

$$u_2 = \cancel{\left(\int_1^5 \frac{x}{2} dx \right)} \cdot \frac{1}{2} = \frac{6}{2} = 3$$

$$\frac{v_2 - u_1 u_2}{\|v_2 - u_1 u_2\|} = \frac{x - 3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{\sqrt{3}}{4} \cdot (x-3)$$

$$u_3 = \left(\int_1^5 \frac{x^2}{2} dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{3}}{4} \cdot (x-3) \cdot x dx \right) \frac{\sqrt{3}}{4} \cdot (x-3) = x^2 - 6x + \frac{23}{3}$$

$$u_3 = \frac{v_3 - u_1 u_3}{\|v_3 - u_1 u_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 \left(x^2 - 6x + \frac{23}{3}\right)^2 dx}} = \frac{3\sqrt{5}}{16} \cdot \left(x^2 - 6x + \frac{23}{3}\right) =$$
$$= \frac{\sqrt{5}}{16} \cancel{3x^2 - 18x + 23}$$

$$B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4} (x-3), \frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \right)$$

(b) označme φ ~~projek~~ ortogonální projekci $\sin x$ ~~do~~ na $\mathbb{R}$ $\text{span}\{1, x, x^2\}$

$$\varphi = \left(\int_1^5 \frac{\sin x}{2} dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x dx \right) \frac{\sqrt{3}}{4} (x-3) +$$

$$+ \left(\int_1^5 \frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \sin x dx \right) \cdot \frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \approx$$

$$\approx \cancel{14} - 0,0525x^2 - 0,322x + 7,6013$$

(c)

$$\|\sin x - \varphi\| = \int_1^5 (\sin x + 0,0525x^2 + 0,322x - 7,6013)^2 dx$$

$$\approx 0,3188$$